

Degenerate Principal Series Functional-Equation Wall

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Summary

Pass 81 interprets the Pass-80 Borel as a maximally degenerate principal series. Since the solid symplectic group has collapsed to

$$\mathrm{Sp}(H) = B = \mathbb{Q}^\times \ltimes \epsilon,$$

the flag variety is a point and

$$I(s) = \mathrm{Ind}_B^{\mathrm{Sp}(H)} \chi_s \cong \chi_s.$$

There is no opposite unipotent cell and no standard intertwiner realizing $s \leftrightarrow -s$.

Main Claims

- **Flag collapse.** The finite groups still have nontrivial projective lines and Bruhat cells, but the solid limit has $\bar{U} = \mathrm{Hom}(\epsilon, \mathbb{Q}) = 0$.
- **No functional equation.** The standard intertwiner $M(w, s) : I(s) \rightarrow I(-s)$ would integrate over $\bar{U}$. Since $\bar{U} = 0$ as the missing cross-polarization morphism, the solid representation has no nonzero Weyl-mediated functional equation.
- **Finite/limit dichotomy.** At every finite level the DFT conjugates dilation by t to dilation by t^{-1} and Gauss sums supply the local c -factor. The symmetry is destroyed only in the solid limit.

Machine Verification

`code/scripts/check-pass81.py` produced `artifacts/reports/pass81-degenerate-principal-series-functional-equation-check.json` with overall PASS. It checks finite flag sizes, the DFT dilation identity, Gauss-sum norms for primes through 23, and the limit obstruction separating the vanished flip tower from the nonzero shear tower.

Next Use

The remaining automorphic residue is not a functional equation but a possible constant-term or Whittaker-style functional. That is the question taken up in Pass 82.